

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405807
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Paulraj; Deepaganesh et al.

System and method for boot and shutdown synchronization of server operating system and component operating system

Abstract

In one or more embodiments, a server may have a server operating system (OS) and a component with a component OS. During a boot process, embodiments may ensure the server OS does not boot until after the component OS boots. During a shutdown process, embodiments may ensure the server OS does not shut down until after the component OS shuts down. In systems with multiple Remote Access Controller, embodiments may synchronize processing between the two RACs.

Inventors: Paulraj; Deepaganesh (Bangalore, IN), Ballard; Lee E. (Georgetown, TX)

Applicant: Dell Products L.P. (Round Rock, TX)

Family ID: 94372020

Assignee: Dell Products L.P. (Round Rock, TX)

Appl. No.: 18/359555

Filed: July 26, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250036421 A1	Jan. 30, 2025

Publication Classification

Int. Cl.: G06F9/4401 (20180101)

U.S. Cl.:

CPC G06F9/4406 (20130101);

Field of Classification Search

CPC: G06F (9/4406)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5577210	12/1995	Abdous	711/111	G06F 9/4416
5794031	12/1997	Nakadai	713/2	G06F 9/442
5878248	12/1998	Tehrani	709/200	G06F 3/1454
6021262	12/1999	Cote	709/224	G06F 11/0781
6094688	12/1999	Mellen-Garnett	718/101	G06F 9/54
10509758	12/2018	Habusha	N/A	G06F 13/4022
11080168	12/2020	Choudhury	N/A	G06F 11/3466
2003/0097553	12/2002	Frye, Jr.	713/2	H04L 67/34
2003/0163680	12/2002	Wang	713/1	H04L 67/34
2004/0024853	12/2003	Cates	718/105	H04L 67/1001
2004/0148329	12/2003	Ogasawara	709/200	H04L 67/1097
2005/0101331	12/2004	Tamaki	455/456.1	H04W 64/00
2008/0278285	12/2007	Matsushima	340/5.74	G11B 20/00695
2009/0118018	12/2008	Perlman	463/42	H04N 21/2143
2010/0235782	12/2009	Powell	715/764	G16H 40/67
2011/0047540	12/2010	Williams	715/740	G06F 8/61
2011/0214176	12/2010	Burch	718/1	G06F 9/45533
2013/0346576	12/2012	Huang	709/223	H04L 61/4511
2014/0006815	12/2013	Castro-Leon	713/310	G06F 9/5094
2015/0012736	12/2014	Sun	713/2	G06F 9/4401
2015/0074659	12/2014	Madsen	717/177	G06F 9/45558
2015/0227400	12/2014	Yamashita	719/318	G06F 12/0292
2015/0271297	12/2014	Zimmer	709/203	H04L 1/0041
2016/0349817	12/2015	Kumar	N/A	G06F 3/0647
2019/0245918	12/2018	Xu	N/A	H04L 63/0869
2020/0051419	12/2019	Malaver	N/A	H04L 67/10
2024/0403435	12/2023	Moore	N/A	G06F 21/575

OTHER PUBLICATIONS

Integrated Dell Remote Access Controller User's Guide version 9. Dec. 2020 (Year: 2020). cited by examiner

Primary Examiner: Myers; Paul R.

Attorney, Agent or Firm: McDermott Will & Schulte LLP

Background/Summary

BACKGROUND

Field of the Disclosure

(1) This disclosure relates generally to information handling systems and remote access controllers found in servers, and more particularly to systems and methods for synchronizing boot and shutdown processes.

Description of the Related Art

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

SUMMARY

(3) Embodiments may be directed to a system comprising a server and a private cloud server (PCS) communicatively coupled to the server. The server may comprise a plurality of components, each component storing a component operating system (OS); a server memory storing: a server OS; a component node for each component OS; and a remote access controller (RAC). The PCS may comprise a PCS processor and a PCS memory storing: a boot synchronization service executable by the PCS processor to: communicate with the RAC to identify the plurality of components; determine a boot order for the plurality of components; and communicate with the RAC to complete, based on the boot order, a boot process for the plurality of components before completing a boot process for the server OS. The PCS may comprise a shutdown synchronization service configured to: determine a shutdown order for the plurality of components; and communicate with the RAC to complete, based on the shutdown order, a shutdown process for the plurality of components before completing a shutdown process for the server OS.

(4) In some embodiments, the PCS memory stores: a policy manager service storing a set of policies; and a boot order/timeout configuration service configured to determine the boot order based on a policy stored in the policy manager service, wherein the boot synchronization service is configured to communicate with the boot order/timeout configuration service to determine the boot order. In some embodiments, the plurality of components comprises a data processing unit (DPU); and the PCS memory stores a DPU OS halt service executable by the PCS processor to communicate with a DPU Basic Input/Output System (BIOS)/OS node to determine when the boot process for the DPU OS is completed. In some embodiments, the PCS memory stores a DPU monitoring service executable by the PCS processor to halt the shutdown process for the server OS until the shutdown process for the plurality of components is completed. In some embodiments, the PCS memory stores a server OS halt service executable by the PCS processor to communicate with a server Basic Input/Output System (BIOS) node to halt the boot process for the server OS until the boot process for the plurality of components is completed. In some embodiments, the PCS memory stores a complex programmable logic device (CPLD) monitoring service executable by the PCS processor to maintain a power supply to the server until the boot process for the plurality of components is completed.

(5) Embodiments may be directed to a data center comprising a plurality of servers, each server comprising: a plurality of components, each component storing a component operating system (OS); a server memory storing: a server OS; a component node for each component OS; and a remote access controller (RAC). The data center may further comprise a private cloud server (PCS) communicatively coupled to the plurality of servers, the private cloud server comprising: a PCS processor; and a PCS memory storing: a boot synchronization service executable by the PCS processor to: communicate with the RAC in a server of the plurality of servers to identify the plurality of components; determine a boot order for the plurality of components; and communicate with the RAC to complete, based on the boot order, a boot process for the plurality of components before completing a boot process for the server OS; a shutdown synchronization service configured to: determine a shutdown order for the plurality of components; and communicate with the RAC to complete, based on the shutdown order, a shutdown process for the plurality of components before completing a shutdown process for the server OS.

(6) In some embodiments, the PCS memory stores: a policy manager service storing a set of policies; and a boot order/timeout configuration service configured to determine the boot order based on a policy stored in the policy manager service, wherein the boot synchronization service is configured to communicate with the boot order/timeout configuration service to determine the boot order. In some embodiments, the plurality of components comprises a data processing unit (DPU); and the PCS memory stores a DPU OS halt service executable by the PCS processor to communicate with a DPU Basic Input/Output System (BIOS)/OS node to determine when the boot process for the DPU OS is completed. In some embodiments, the PCS memory stores a DPU monitoring service executable by the PCS processor to halt the shutdown process for the server OS until the shutdown process for the plurality of components is completed. In some embodiments, the PCS memory stores a server OS halt service executable by the PCS processor to communicate with a server Basic Input/Output System (BIOS) node to halt the boot process for the server OS until the boot process for the plurality of components is completed. In some embodiments, the PCS memory stores a complex programmable logic device (CPLD) monitoring service executable by the PCS processor to maintain a power supply to the server until the boot process for the plurality of components is completed.

(7) Embodiments may be directed to a method comprising: detecting, by a private cloud server (PCS) processor in a server, an indication to boot the server; communicating, with a set of components in the server, an indication that power is being received, wherein each component in the set of components has a component operating system (OS); communicating with a server Basic Input/Output System (BIOS)/operating system (OS)/witness node, an indication to cause the server BIOS/OS/witness node to wait; receiving, from the set of components, an indication that each component in the set of components has finished a component boot process; and communicating a signal to the server BIOS/OS/witness node to enable the server BIOS/OS/witness node to finish a server boot process.

(8) In some embodiments, the method comprises storing a set of policies in a policy manager service; and determining, by a boot order/timeout configuration service, the boot order based on a policy stored in the policy manager service, wherein the boot synchronization service is configured to communicate with the boot order/timeout configuration service to determine the boot order. In some embodiments, the plurality of components comprises a data processing unit (DPU); and the method comprises communicating, by a DPU OS halt service stored in the PCS memory, with a DPU Basic Input/Output System (BIOS)/OS node to determine when the boot process for the DPU OS is completed. In some embodiments, the method comprises a DPU monitoring service halting the shutdown process for the server OS until the shutdown process for the plurality of components is completed. In some embodiments, the method comprises a server OS halt service halting the boot process for the server OS until the boot process for the plurality of components is completed. In some embodiments, the method comprises a complex programmable logic device (CPLD)

monitoring service maintaining a power supply to the server until the boot process for the plurality of components is completed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the present disclosure and its features/advantages, reference is now made to the following description, taken in conjunction with the accompanying drawings, which are not drawn to scale, and in which:
- (2) FIG. 1 depicts an example architecture of a plurality of data centers, with each data center containing a plurality of servers, with each server communicatively coupled to an external cloud server for managing the server;
- (3) FIG. 2 depicts an example of a server with a Remote Access Controller (RAC), illustrating a set of services commonly installed in firmware on a RAC for remotely monitoring and managing the server;
- (4) FIG. 3 depicts a system architecture of a plurality of data centers communicatively coupled to an external cloud server, with each data center containing a plurality of servers and a private cloud server, with each server being communicatively coupled to the private cloud server for managing the servers, in accordance with some embodiments;
- (5) FIG. 4 depicts a portion of the system architecture of FIG. 3, illustrating an embodiment of a RAC that may be installed in a server and communicatively coupled to a private cloud server and one or more external cloud servers for supporting the server;
- (6) FIG. 5 depicts a portion of the system architecture of FIG. 3, illustrating one embodiment of a private cloud server with firmware installed and communicatively coupled to a RAC in a server and one or more external cloud servers;
- (7) FIG. 6 depicts one embodiment of a system with a centralized repository for managing configurations of a server;
- (8) FIG. 7 depicts one embodiment of a portion of an information handling system having two baseboard management controllers (BMCs) communicating with each other through a Complex Programmable Logic Devices (CPLDs) and over a bus; and
- (9) FIG. 8 depicts a flow diagram illustrating a method for synchronized booting of an information handling system in accordance with some embodiments; and
- (10) FIG. 9 depicts a flow diagram illustrating a method for synchronized shutdown of an information handling system in accordance with some embodiments.

DETAILED DESCRIPTION

- (11) In the following description, details are set forth by way of example to facilitate discussion of the disclosed subject matter. It should be apparent to a person of ordinary skill in the field, however, that the disclosed embodiments are examples and not exhaustive of all possible embodiments.
- (12) As used herein, a reference numeral refers to a class or type of entity, and any letter following such reference numeral refers to a specific instance of a particular entity of that class or type. Thus, for example, a hypothetical entity referenced by '12A' may refer to a particular instance of a particular class/type, and the reference '12' may refer to a collection of instances belonging to that particular class/type or any one instance of that class/type in general.
- (13) An information handling system (IHS) may include a hardware resource or an aggregate of hardware resources operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, and/or utilize various forms of information, intelligence, or data for business, scientific, control, entertainment, or other purposes, according to one or more embodiments. For example, an IHS may be a personal computer, a

desktop computer system, a laptop computer system, a server computer system, a mobile device, a tablet computing device, a personal digital assistant (PDA), a consumer electronic device, an electronic music player, an electronic camera, an electronic video player, a wireless access point, a network storage device, or another suitable device and may vary in size, shape, performance, functionality, and price. In one or more embodiments, a portable IHS may include or have a form factor of that of or similar to one or more of a laptop, a notebook, a telephone, a tablet, and a PDA, among others. For example, a portable IHS may be readily carried and/or transported by a user (e.g., a person). In one or more embodiments, components of an IHS may include one or more storage devices, one or more communications ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, and a video display, among others. In one or more embodiments, IHS may include one or more buses operable to transmit communication between or among two or more hardware components. In one example, a bus of an IHS may include one or more of a memory bus, a peripheral bus, and a local bus, among others. In another example, a bus of an IHS may include one or more of a Micro Channel Architecture (MCA) bus, an Industry Standard Architecture (ISA) bus, an Enhanced ISA (EISA) bus, a Peripheral Component Interconnect (PCI) bus, HyperTransport (HT) bus, an inter-integrated circuit (I2C) bus, a serial peripheral interface (SPI) bus, a low pin count (LPC) bus, an enhanced serial peripheral interface (eSPI) bus, a universal serial bus (USB), a system management bus (SMBus), and a Video Electronics Standards Association (VESA) local bus, among others.

(14) In one or more embodiments, an IHS may include firmware that controls and/or communicates with one or more hard drives, network circuitry, one or more memory devices, one or more I/O devices, and/or one or more other peripheral devices. For example, firmware may include software embedded in an IHS component utilized to perform tasks. In one or more embodiments, firmware may be stored in non-volatile memory, such as storage that does not lose stored data upon loss of power. In one example, firmware associated with an IHS component may be stored in non-volatile memory that is accessible to one or more IHS components. In another example, firmware associated with an IHS component may be stored in non-volatile memory that may be dedicated to and includes part of that component. For instance, an embedded controller may include firmware that may be stored via non-volatile memory that may be dedicated to and includes part of the embedded controller.

(15) An IHS may include a processor, a volatile memory medium, non-volatile memory media, an I/O subsystem, and a network interface. Volatile memory medium, non-volatile memory media, I/O subsystem, and network interface may be communicatively coupled to processor. In one or more embodiments, one or more of volatile memory medium, non-volatile memory media, I/O subsystem, and network interface may be communicatively coupled to processor via one or more buses, one or more switches, and/or one or more root complexes, among others. In one example, one or more of a volatile memory medium, non-volatile memory media, an I/O subsystem, and a network interface may be communicatively coupled to the processor via one or more PCI-Express (PCIe) root complexes. In another example, one or more of an I/O subsystem and a network interface may be communicatively coupled to processor via one or more PCIe switches.

(16) In one or more embodiments, the term “memory medium” may mean a “storage device”, a “memory”, a “memory device”, a “tangible computer readable storage medium”, and/or a “computer-readable medium”. For example, computer-readable media may include, without limitation, storage media such as a direct access storage device (e.g., a hard disk drive, a floppy disk, etc.), a sequential access storage device (e.g., a tape disk drive), a compact disk (CD), a CD-ROM, a digital versatile disc (DVD), a random access memory (RAM), a read-only memory (ROM), a one-time programmable (OTP) memory, an electrically erasable programmable read-only memory (EEPROM), and/or a flash memory, a solid state drive (SSD), or any combination of the foregoing, among others.

(17) In one or more embodiments, one or more protocols may be utilized in transferring data to

and/or from a memory medium. For example, the one or more protocols may include one or more of small computer system interface (SCSI), Serial Attached SCSI (SAS) or another transport that operates with the SCSI protocol, advanced technology attachment (ATA), serial ATA (SATA), a USB interface, an Institute of Electrical and Electronics Engineers (IEEE) 1394 interface, a Thunderbolt interface, an advanced technology attachment packet interface (ATAPI), serial storage architecture (SSA), integrated drive electronics (IDE), or any combination thereof, among others.

(18) A volatile memory medium may include volatile storage such as, for example, RAM, DRAM (dynamic RAM), EDO RAM (extended data out RAM), SRAM (static RAM), etc. One or more of non-volatile memory media may include nonvolatile storage such as, for example, a read only memory (ROM), a programmable ROM (PROM), an erasable PROM (EPROM), an electrically erasable PROM, NVRAM (non-volatile RAM), ferroelectric RAM (FRAM), a magnetic medium (e.g., a hard drive, a floppy disk, a magnetic tape, etc.), optical storage (e.g., a CD, a DVD, a BLU-RAY disc, etc.), flash memory, a SSD, etc. In one or more embodiments, a memory medium can include one or more volatile storages and/or one or more nonvolatile storages.

(19) In one or more embodiments, a network interface may be utilized in communicating with one or more networks and/or one or more other information handling systems. In one example, network interface may enable an IHS to communicate via a network utilizing a suitable transmission protocol and/or standard. In a second example, a network interface may be coupled to a wired network. In a third example, a network interface may be coupled to an optical network. In another example, a network interface may be coupled to a wireless network. In one instance, the wireless network may include a cellular telephone network. In a second instance, the wireless network may include a satellite telephone network. In another instance, the wireless network may include a wireless Ethernet network (e.g., a Wi-Fi network, an IEEE 802.11 network, etc.).

(20) In one or more embodiments, a network interface may be communicatively coupled via a network to a network storage resource. For example, the network may be implemented as, or may be a part of, a storage area network (SAN), personal area network (PAN), local area network (LAN), a metropolitan area network (MAN), a wide area network (WAN), a wireless local area network (WLAN), a virtual private network (VPN), an intranet, an Internet or another appropriate architecture or system that facilitates the communication of signals, data and/or messages (generally referred to as data). For instance, the network may transmit data utilizing a desired storage and/or communication protocol, including one or more of Fibre Channel, Frame Relay, Asynchronous Transfer Mode (ATM), Internet protocol (IP), other packet-based protocol, Internet SCSI (iSCSI), or any combination thereof, among others.

(21) In one or more embodiments, a processor may execute processor instructions in implementing at least a portion of one or more systems, at least a portion of one or more flowcharts, at least a portion of one or more methods, and/or at least a portion of one or more processes. In one example, a processor may execute processor instructions from one or more memory media in implementing at least a portion of one or more systems, at least a portion of one or more flowcharts, at least a portion of one or more methods, and/or at least a portion of one or more processes. In another example, a processor may execute processor instructions via a network interface in implementing at least a portion of one or more systems, at least a portion of one or more flowcharts, at least a portion of one or more methods, and/or at least a portion of one or more processes.

(22) In one or more embodiments, a processor may include one or more of a system, a device, and an apparatus operable to interpret and/or execute program instructions and/or process data, among others, and may include one or more of a microprocessor, a microcontroller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), and another digital or analog circuitry configured to interpret and/or execute program instructions and/or process data, among others. In one example, a processor may interpret and/or execute program instructions and/or process data stored locally (e.g., via memory media and/or another component of an IHS). In another example, a processor may interpret and/or execute program instructions and/or process data

stored remotely (e.g., via a network storage resource).

(23) In one or more embodiments, an I/O subsystem may represent a variety of communication interfaces, graphics interfaces, video interfaces, user input interfaces, and/or peripheral interfaces, among others. For example, an I/O subsystem may include one or more of a touch panel and a display adapter, among others. For instance, a touch panel may include circuitry that enables touch functionality in conjunction with a display that is driven by a display adapter.

(24) A non-volatile memory medium may include an operating system (OS) and applications (APPs). In one or more embodiments, one or more of an OS and APPs may include processor instructions executable by a processor. In one example, a processor may execute processor instructions of one or more of OS and APPs via a non-volatile memory medium. In another example, one or more portions of the processor instructions of one or more of an OS and APPs may be transferred to a volatile memory medium and a processor may execute the one or more portions of the processor instructions.

(25) Non-volatile memory medium may include information handling system firmware (IHSFW). In one or more embodiments, IHSFW may include processor instructions executable by a processor. For example, IHSFW may include one or more structures and/or one or more functionalities of and/or compliant with one or more of a basic input/output system (BIOS), an Extensible Firmware Interface (EFI), a Unified Extensible Firmware Interface (UEFI), and an Advanced Configuration and Power Interface (ACPI), among others. In one instance, a processor may execute processor instructions of IHSFW via non-volatile memory medium. In another instance, one or more portions of the processor instructions of IHSFW may be transferred to volatile memory medium, and processor may execute the one or more portions of the processor instructions of IHSFW via volatile memory medium.

(26) Data centers may have large numbers of information handling systems such as servers for processing information. A data center facility may have one or more floors with each floor having racks of servers. A server may be processing a set of information independently or a group of servers may be working on the same set of information.

(27) Turning now to FIG. 1, an architectural diagram of a plurality of data centers **10** illustrates that each data center **10** may contain a plurality of information handling systems **12** (also referred to as servers **12**). Data center **10-1** may contain M servers **12** and data center **10-2** may contain N servers, where each of M and N may be any number between two to over a thousand. Each server **12** may comprise Remote Access Controller (RAC) **14** to allow remote monitoring and management of any server **12** in data center **10**. RAC **14** may comprise a Dell Remote Access Controller (DRAC) or an integrated Dell Remote Access Controller (iDRAC) for remote monitoring and management.

(28) Turning to FIG. 2, each server **12** has remote access controller (RAC) **14** with firmware to allow for remote monitoring and managing of server **12**, and server **12** may need to communicate with one or more other servers **12** in data center **10**. Firmware within RAC **14** may include a Representational State Transfer Application Programming Interface (REST API) **16** that allows servers **12** to communicate with each other. Remote monitoring may require communications with processes running in RAC **14**. A D-Bus **18** or other inter-process communications mechanism may enable local communication between processes stored in firmware on RAC **14**.

(29) Firmware in RAC **14** may include a set of built in features **30** executable by a processor in RAC **14**. For example, each server **12** has a power supply unit (PSU) for receiving power from a power source and transforming the power into usable power by the server **12**. Firmware in RAC **14** includes power control **30-1** as a feature **30** for monitoring of power received by the PSU to determine when server **12** has switched power modes and allow for remote switching of server **12** between power modes. Similarly, server **12** has a fan for cooling server **12** and firmware in RAC **14** includes thermal control **30-2** as a feature **30** for monitoring of temperatures in server **12** to determine if the fan is operating, determining when to operate the fan and enabling remote

operation of the fan.

(30) Firmware in RAC **14** may also store hardware abstraction layer (HAL) **22** for communication with and monitoring of peripheral devices. Within HAL **22**, features such as Inter-Integrated Circuit (I2C) protocol **22-1**, Improved Inter-Integrated Circuit (**13C**) protocol **22-2**, Serial Peripheral Interface (SPI) **22-3**, Reduced Media-Independent Interface (RMII) **22-4**, Peripheral Connect Interface Express Vendor Defined Message (PCIe VDM) **22-5** and management component transport protocol (MCTP) **22-6** allow communications with chips, processors and microcontrollers in server **12**.

(31) Server **12** includes hardware **24** such as processors for processing information, memory for storing information, a fan for cooling server **12** including devices and components in server **12**, a network integrated circuit (NIC) for communications, other controllers such as RAID controllers, and Complex Logic Programmable Devices (CPLDs). Accordingly, Firmware in RAC **14** may include other features such as Basic Input Output Service (BIOS) **26-1**, NIC **26-2**, fan control **26-3**, PSU control **26-4** for operating a PSU, RAID feature **26-5** for managing a RAID and CPLD feature **26-6** for monitoring and managing a CPLD.

(32) Firmware in RAC **14** may typically include other features **30** for monitoring and managing servers **12** in data centers **10**. The examples depicted in FIG. **3** and provided below represent more common features found in RAC **14**, and RAC **14** may store additional features **30** not listed.

(33) Web Graphical User Interface (GUI) **30-1** is a web-based application that processes network events for displaying the data in a graphical format for users to view on an attached display. Redfish® **30-2** is an application programming interface (API) that uses RESTful semantics to access data defined in model format for systems management. Remote Access Controller admin (RACADM) feature **30-3** is a scriptable interface to allow remote configuration of RAC **14**. Simple Network Management Protocol (SNMP) feature **30-4** may be used to collect data related to network changes and determine the status of network-connected devices. Hardware inventory feature **30-5** may maintain an inventory and properties of all hardware installed on server **12**. Software inventory feature **30-6** may maintain an inventory and versions of all software running on server **12**. Firmware update feature **30-7** may maintain a list of firmware including versions and facilitate updates of any firmware on server **12**. Monitoring feature **30-8** may monitor operation of components or devices in server **12** and record values or may define what operations are to be monitored and how the monitoring should occur. RAC **14** may include database **32** for storing information about components or devices in server **12**. Network Controller-Sideband Interface (e.g., NC-SI 1.2) feature **30-10** defines a control communication protocol between a baseboard management controller (BMC) and one or more Network Interface Controllers (NICs). Platform Level Data Model (PLDM) feature **30-11** defines the contents of a firmware update package. Security Protocol and Data Models (SPDM) feature **30-12** enables authentication, attestation and key exchange for enabling security of server **12**. Management Control Transport Protocol (MCTP) feature **30-13** stores message formats, transport descriptions, message exchange patterns and endpoint characteristics related to communications between components.

(34) Servers **12** described above have several shortcomings. All features **30** are stored in memory in each RAC **14** of each server **12**, regardless of whether a feature **30** will be used for a particular server **12**. Furthermore, features **30** are getting more robust and require more memory to manage devices in servers **12**. For example, RAC **14** in some generations of servers **12** may have 512 Megabytes (MB) of memory to store all features **30**, RAC **14** in later generations of servers **12** may have 1 Gigabyte (GB) of memory to store all features **30**, and RAC **14** in later generations of servers **12** may have 2 GB of memory to store all features **30**. Although each server **12** is provided with the full suite of features **30**, many customers or servers **12** require a limited set of features **30**. For security reasons, some users prefer to have only firmware code which they know they will use. Subscription-based licensing, in which a user pays for only the features needed, may be preferred. However, only about 35% of datacenter servers **12** contain the latest firmware for RAC **14**, and

features **30** are often unnecessarily tied to specific hardware capabilities. Customization of features **30** via Software Developer Kits (SDK) may be limited or unavailable due to hardware capabilities or software requirements. Some devices (e.g., Baseboard Management Controllers (BMCs)) are expected to be commoditized, resulting in greater complexity and/or more features **30** that would need to be installed on each RAC **14**.

(35) Turning to FIGS. **3**, **4** and **5**, embodiments disclosed herein allow a remote access controller (RAC) **314** in server **312** to have firmware that has a minimum number of features **30** that can be augmented as needed with any number of available services **350** executing on a private cloud server **360**. In some embodiments, RAC **314** may be manufactured with firmware including a minimum Hardware Abstraction Layer (HAL). Some controls **20** and features **30** may remain as built-in controls **20** or features **30** in firmware in RAC **314**. For example, any time server **12** is operating, PSU control **20-1** monitors power supplied to the PSU and is therefore maintained in firmware in RAC **314**. Embodiments may augment built-in features **30** with available services **350** that can be retrieved from external cloud server **210** in cloud **4** and installed and executed on private cloud server **360** or possibly installed on RAC **314**, discussed in greater detail below. If needed by RAC **314**, available services **350** may be delivered/deployed by leveraging and extending on top of existing firmware in RAC **314**.

(36) Advantageously, embodiments may provide vendor-agnostic, unified and seamless system management across generations of servers **312**. Embodiments may further provide accelerated delivery of new available services **350** with subscription-based granular licensing.

(37) Referring to FIG. **3**, data centers **310** may each have multiple servers **312**. Data center **310-1** may have a first plurality of servers **312** (e.g., **2** to **M**) divided into **P** groups **340**, wherein group **340-1** may have a first set of servers **312** (e.g., **1** to **A**), group **340-2** may have a second set of servers **312** (e.g., **1** to **B**) and group **340-P** may have a third set of servers **312** (e.g., **1** to **C**). Data center **310-1** may further comprise private cloud server **360-1** communicatively coupled to all servers **312-1** to **312-M** in data center **310-1**. Private cloud server **360-1** may be communicatively coupled to external cloud server **210** in cloud **4**.

(38) Data center **310-2** may have a second plurality of servers **312** (e.g., **2** to **N**) divided into **Q** groups **340**, wherein group **340-4** may have a first set of servers **312** (e.g., **1** to **D**), group **340-2** may have a second set of servers **312** (e.g., **1** to **E**) and group **340-Q** may have a third set of servers **312** (e.g., **1** to **F**). Data center **310-2** may further comprise private cloud server **360-2** communicatively coupled to all servers **312-1** to **312-N** in data center **310-2**. Private cloud server **360-2** may be communicatively coupled to external cloud servers **210** in cloud **4**.

(39) Groups **340** may be logical groups, discussed in greater detail with respect to FIG. **6**.

(40) Referring to FIG. **4**, external cloud server **210** may communicate with private cloud servers **360** over a network to provide available services **350**.

(41) Cloud integration service **210-1** may include data monitoring **214** and cloud services **216**. Dell CloudIQ is an example of a data monitoring service **214** that may monitor operation of servers **312** in data centers **310** for data protection. Dell APEX is an example catalog of cloud services **216** that may integrate infrastructure and software for cloud operations.

(42) Infrastructure Solutions Group (ISG) server **210-2** may provide storage and networking services, and may include third-party services **350-18**. In some embodiments, ISG server **210-2** may provide RAC proxy on Data Processing Unit (DPU) service **220** to private cloud server **360** to offload processing.

(43) Management Service Catalog (MSC) server **210-3** may store a database of available services **350** that can be provided to private cloud servers **360**. In some embodiments, private cloud server **360** communicates with MSC server **210-3** to retrieve available services **350**. In some embodiments, MSC server **210-3** communicates with private cloud server **360** to install available services **350**.

(44) IOT integration service **210-4** may enable devices to communicate with other devices and

enable external cloud servers **210** to communicate with or monitor devices in servers **312**.

(45) Autonomous server manager **210-5** may provide server management without a user.

(46) Turning to one or more of FIGS. **4** and **5**, servers **312** may be similar to servers **12** depicted in FIG. **2** in that they comprise hardware such as a PSU for receiving power from a source and transforming the power into usable power by the server, a fan, network IC (NIC) for communications, other controllers such as RAID controllers, and Complex Logic Programmable Devices (CPLDs). Servers **312** also include BIOS **26-1**, NIC **26-2**, fan control **26-3**, PSU service **26-4** for operating a PSU, remote controller service **26-5** for managing a RAID and CPLD service **26-6** for monitoring and managing CPLDs.

(47) Firmware within RAC **314** on servers **312** may include a REpresentational State Transfer Application Programming Interface (REST API) **16** that allows servers **312** to communicate with each other. Remote monitoring may require communications with processes running in RAC **314**. A D-Bus **18** or other inter-process communications mechanism enables local communication between processes on RAC **314**. RAC **314** may further comprise Remote Procedure Call (RPC) queue **302**, D-Bus to RPC mapper **304** and D-Bus to REST mapper **306** for available services **350** on private cloud server **360** to communicate with HAL **22**. In some embodiments, RPC mapper **304** may be a Google RPC (gRPC) mapper **304**.

(48) RAC **314** may also store hardware abstraction layer (HAL) **22** for communication and monitoring of peripheral devices. Within HAL **22**, I2C **22-1**, I3C **22-2**, SPI **22-3**, RMII **22-4**, PCIe VDM **22-5** and MCTP **22-6** allow communications with chips, processors and microcontrollers in server **312**.

(49) Available Services May be Stored and Executed on a Private Cloud Server

(50) Embodiments of private cloud server **360** may store a set of available services **350** retrieved from MSC server **210-3**. In some embodiments, private cloud server **360** may determine, based on one or more of an application to be executed on a server **312** or a data set to be processed by a server **312**, that a particular available service is needed and communicate with MSC server **210-3** to retrieve the available service **350**. In some embodiments, private cloud server **360** may communicate with MSC server **210-3** to retrieve an available service **350** based on a subscription. Available services **350** may be executed by private cloud server **360** unless there is a need to have RAC **314** execute the available service **350**.

(51) Available Services May be Installed in a RAC if Needed

(52) Embodiments may install available services **350** in RAC **314** from private cloud server **360**, wherein the set of available services **350** stored in RAC **314** may be less than the total number of available services **350** stored in private cloud server **360** and less than the plurality of available services **350** stored in MSC server **210-3**. Available services **350** may be installed in RAC **314** based on one or more of performance or capability. Once the process is complete or the available service **350** is no longer time-sensitive, embodiments may uninstall the available service **350** from RAC **314**.

(53) As a device-based example, referring to available service **350-13** (i.e., MCTP **350-13**) and available service **350-10** (i.e., NC-SI 1.2 protocols **350-10**), if a process sends a request to CPLD **26-6**, a response may be expected in a minimum response time of about 50 milliseconds (ms) and a maximum response time of about 4 seconds. If the MCTP **350-13** and NC-SI 1.2 protocols **350-10** are stored in private cloud server **360**, a response may take 5 seconds. Embodiments may install NC-SI 1.2 protocols **350-10** as a time sensitive plug-in **340** to ensure a response is received in less than 4 seconds. Once NC-SI 1.2 protocols **350-10** is not needed, NC-SI 1.2 protocols **350-10** may be uninstalled from RAC **314**.

(54) As another example, referring to available service **350-11** (i.e., PLDM **350-11**) and available service **350-7** (i.e., Firmware update service **350-7**), if there is a firmware update and PLDM **350-11** executing on private cloud server **360** cannot deliver the update payload within a maximum time (e.g., 4 seconds), the firmware update may time out. In this case, PLDM **350-11** and Firmware

update service **350-7** may be installed as time sensitive plug-ins **340** to ensure the firmware update payload can be delivered in time. Once the firmware update payload is delivered, PLDM **350-11** and Firmware update service **350-7** may be uninstalled from RAC **314**.

(55) Existing Available Services Stored on Private Cloud Server

(56) Box **352** contains a non-limiting list of available services **350** that may be installed on private cloud server **360** to communicate with RACs **314** on any servers **312** in data centers **310**, wherein available services **350** in box **352** may function similar to features **30** described above with respect to FIG. **2** but execute on private cloud server **360**. For example, RAC **314** may be configured with a minimum number of features **30** and private cloud server **360** may store and execute Web GUI service **350-1**; Redfish® server management service **350-2**; RACADM command-line service **250-3**; SNMP service **350-4**; Hardware (HW) inventory service **350-5** for tracking information and properties of hardware installed on servers **312**; Firmware (FW) inventory service **350-6** for tracking information and versions of firmware installed on servers **312**; FW update service **350-7** for installing Firmware updates in servers **312**; Monitoring service **350-8** for coordinating monitoring of operation of servers **312**; Unified database **350-9** for storing information for sharing among processes and applications executing on servers **312**; NC-SI service **350-10**; PLDM service **350-11**; SPDM service **350-12** and MCTP service **350-13**. In some embodiments, available services **350** may be retrieved from Management Service Catalog server **210-3** through subscriptions, wherein any available service **350** may be retrieved as needed.

(57) New Services and Services with Increased Scope

(58) Embodiments may allow private cloud server **360** to provide additional available services **350** and available services **350** with increased scope that increase the capabilities of RAC **314**. Box **354** contains a non-limiting list of available services **350** that may be retrieved from MSC server **210-3** and installed on private cloud server **360**. Some available services **350** in box **354** may be executed by private cloud server **360**. Some available services **350** in box **354** may be installed on RAC **314** on any server **312** in data center **310**.

(59) Credentials Service

(60) A user must have administrative credentials to run RACADM commands remotely. When a user wants to run RACADM commands, the user must first be validated by RAC **314**.

Traditionally, RAC **14** corresponding to server **12** in FIG. **2** may store credentials for one to about sixteen users. Embodiments described herein allow private cloud server **360** to download and execute credentials service **350-15**, which may be configured to manage credentials for one user to more than sixteen users and map the credentials to RACs **314** as needed. In some embodiments, unified database **350-9** on private cloud server **360** may be configured to store credentials for a large number of users (e.g., over a thousand) to increase the scope of credentials service **350-9** over credentialing features **30** commonly found in RACs **14**. In some embodiments, private cloud server **360** may be configured to install credential service **350-15** and store a database in unified database **350-9** with the credentials for any number of users between one to over a thousand. As the number of users changes, private cloud server **360** may update the database. In some embodiments, credentials service **350-15** may validate users. In some embodiments, credentials service **350-15** may register itself as a credentials manager, receive requests, and validate users such that RAC **314** does not validate but knows the user is validated.

(61) Telemetry Service

(62) Some Baseboard Management Controllers (BMCs) have no intelligence to adjust telemetry collection based on server events/errors or have different streaming rate for each metric.

Traditionally, RAC **14** in server **12** may send the same information multiple times or at different rates, tying up networking resources and memory. Telemetry service **350-16** may be retrieved from MSC server **210-3** and installed on private cloud server **360**. In some embodiments, an Open Telemetry (OTEL) service **350-16** may be stored in private cloud server **360** as an available service **350**. When servers **312** communicate data for telemetry purposes, telemetry service **350-16**

executing on private cloud server **360** may aggregate data, remove redundant data, or otherwise coordinate the communication of data, resulting in reduced network congestion. In some embodiments, telemetry service **350-16** may be installed in RAC **314** to meet telemetry requirements and then uninstalled after server **312** does not need to meet any telemetry requirements. In some embodiments, telemetry service **350-16** may be installed in RAC **314** as a time-sensitive plug-in **340** to provide quicker responses to telemetry requirements and then uninstalled after server **312** does not need quick responses to meet telemetry requirements.

(63) AI/ML Service

(64) Artificial Intelligence (AI)/Machine learning (ML) service **350-17** may include services necessary for AI/ML. If a server **312** (or a set of servers **312**) is needed for AI/ML, AI/ML service **350-17** may be downloaded to private cloud server **360** for coordinating processing by servers **312** for AI/ML processing.

(65) Third Party Services

(66) Third party services **350-18** may include services needed for particular third-party applications. Advantageously, instead of all available services **350** being tied to particular hardware, embodiments may enable third-party services **350-18** to execute on private cloud server **360**, wherein other available services **350**, RPC queue **302**, D-Bus to RPC mapper **304** and D-Bus to REST mapper **306** enable third-party services **350-18** on private cloud server **360** to communicate with HAL **22** in one or more RACs **314**.

(67) Advanced Power/Thermal Service

(68) Advanced power/thermal service **350-19** may be stored on private cloud server **360** and may refer to an available service **350** that can be executed on private cloud server **360** to communicate with RACs **314** on multiple servers **312** to monitor or control power or thermal operations of one or more servers **312**. For example, data center **310** may have multiple floors with hundreds of racks of servers **312**. Each RAC **314** may communicate with sensors inside a server **312** for remotely and individually monitoring temperature of that server **312**. Advanced power/thermal service **350-19** may allow a user to remotely and collectively monitor and manage temperatures for multiple servers **312**, such as all servers **312** processing a set of information, all servers **312** in a rack or all servers **312** on a floor. In some embodiments, advanced power/thermal service **350-19** may be installed in RAC **314** (e.g., as a time-sensitive plug-in **340**) for quicker response to power/thermal requirements and then uninstalled after server **312** does not need to operate under advanced power/thermal requirements.

(69) Secure Zero Touch Provisioning (sZTP) Service

(70) Secure Zero-Touch Provisioning (sZTP) service **350-20** enables a server **312** to securely provision a network device when booting in a default state. If server **312** is expected to require communication over a network, sZTP service **350-20** executing on private cloud server **360** may ensure the network device is securely provisioned. In some embodiments, sZTP service **350-20** may be installed in RAC **314** for booting and uninstalled once server **312** has successfully booted. In some embodiments, sZTP service **350-20** may be installed on private cloud server **360** and installed on servers **312** (e.g., as a time-sensitive plug-in **340**) as needed to ensure network devices are securely provisioned.

(71) ISM Services

(72) Integrated Dell Remote Access Controller (iDRAC) Service Manager service **350-21** may refer to available services **350-21** that may be executed to monitor and manage operation of RAC **314**.

(73) Fido Device Onboard (FDO) Service

(74) Fast IDentity Online (FIDO) Device Onboard (FDO) service **350-22** allows onboarding and trusting of new devices (e.g., RAC **314**) within an environment. For example, RAC **314** may be Linux-based and one or more available services **350** may be Windows-based. FDO service **350-22** may enable servers **312** to execute Linux-based services **350** and Windows-based services **350**.

Data centers **310** may have hundreds or thousands of servers **312**. Devices in servers **312** may be removed and exchanged for newer devices. Instead of a remote user configuring each device and deploying required applications, FDO service **350-22** allows multiple devices in various servers **312** to be configured correctly. In some embodiments, FDO service **350-22** may be installed in RAC **314** for quicker configuring and deployment and uninstalled once server **312** has successfully configured and deployed new devices. In some embodiments, FDO service **350-22** may be installed on private cloud server **360** and installed on servers **312** (e.g., as a time-sensitive plug-in **340**) as needed to ensure new devices are quickly configured and deployed.

(75) Internet of Things (IOT) Integration Service

(76) IOT Integration service **350-23** may be installed on private cloud server **360** to facilitate integration between devices on servers **312**. Advantageously, instead of each server **312** in multiple data centers **310** communicating with IOT integration server **210-4**, private cloud server **360** may perform some of the integration, wherein other available services **350**, RPC queue **302**, D-Bus to RPC mapper **304** and D-Bus to REST mapper **306** enable private cloud server **360** to communicate with HAL **22** in one or more RACs **314** in servers **312** that may have different devices.

(77) Feature Tour

(78) Feature Tour service **350-24** may be installed to guide users through features **30** available to server **312**. Advantageously, instead of each RAC **314** in each server **312** in multiple data centers **310** storing all the information necessary to guide users through only features **30** installed on that server, private cloud server **360** may store all the information and provide information including features **30** and available services **350** available to a particular server **312**. An end user may not know whether a feature is installed on server **312** or is instead an available service **350** accessible by server **312**. In some embodiments, private cloud server **360** may install feature tour service **350-24** with information relevant to a particular server **312** on RAC **314** associated with the particular server **312**.

(79) Infrastructure Programmer Developer Kit (IPDK) Service

(80) IPDK service **350-25** is an open-source, vendor-agnostic framework of drivers and APIs. IPDK service **350-25** may be too large to install on each RAC **314** in each server **312**. IPDK service **350-25** may be installed on private cloud server **360** for infrastructure offload and management.

(81) For each available service **350**, if the available service **350** is needed for a particular server **312**, a version of the available service **350** may be retrieved from external cloud server **210** and executed on private cloud server **360**, wherein RAC **314** contains a minimum number of built-in controls **20** and features **30** to communicate with private cloud server **360**, wherein private cloud server **360** comprises memory and processors for executing the available service **350**. If, at some later time, the available service **350** is not needed on a particular server **312**, the available service **350** may be uninstalled from RAC **314** but a version may still be stored in private cloud server **360** or the available service **350** may be uninstalled from private cloud server **360**.

(82) MSO—Management Service Orchestrator

(83) MSO **356** may coordinate between features and ensure any service **350** is compatible with other available services **350** and hardware in server **312** and that no conflicts exist. For example, regarding telemetry, a manufacturer may have a default telemetry service **20** installed in firmware, but a customer may want to use Open Telemetry (OTel) service **350-16** or a third-party telemetry service **350-18**. If the customer requests another telemetry service, MSO **356** may determine whether the requested telemetry application will work, if an existing telemetry feature needs to be disabled or an existing telemetry service **350-16** or **350-18** needs to be uninstalled.

(84) Large Send Offload (LSO)

(85) Large Send Offload (LSO) service **358** may increase the egress throughput of high-bandwidth network connections. LSO service **358** may be installed on PCS **312**, reducing workload performed by RAC **314**. In some embodiments, LSO **358** enables communication with RAC Proxy on DPU

service **220** to offload large workloads.

(86) Boot and Shutdown Synchronization

(87) Referring to FIG. **6**, embodiments of server **312** may include two RACs **314**, with each RAC **314** containing RAC firmware **316**.

(88) Each RAC firmware **316** may store boot synchronization (sync) client service **602**, shutdown sync client service **604**, operating system (OS) ready client service **606** and enumeration ready client service **608**.

(89) Server **312** may also store server BIOS/OS/witness node **610** (also referred to as server BIOS/OS node **610**) and DPU BIOS/OS node **612** (also referred to as DPU BIOS/OS node **612**). Server BIOS/OS node **610** may include boot sync client node **602**, shutdown sync client service **604**, operating system (OS) ready client service **606** and enumeration ready client service **608**. DPU BIOS/OS node **612** may include boot sync client node **602**, shutdown sync client service **604**, operating system (OS) ready client service **606** and emulation ready client service **614**.

(90) Private cloud server **360** may comprise firmware storing boot synchronization service **620**, DPU OS ready monitoring service **622**, server BIOS post code service **624**, policy manager service **626**, server BIOS enumeration service **628**, boot order/timeout configuration service **630**, shutdown synchronization service **632**, DPU OS halt service **634**, server OS halt service **636**, and CPLD monitoring service **640**. Each of the services depicted in FIG. **6** may be provided as an available service **350** described above with respect to FIGS. **4** and **5**.

(91) A Data Processing Unit (DPU) may be configured to process information as a way to offload some processing by a CPU. When a DPU OS is booted, PCIe enumeration may begin and one or more PCIe functions can be initialized. DPU OS ready monitoring service **622** may be stored in private cloud server **360** and configured to communicate with DPU BIOS/OS node **612** to determine when a DPU OS has finished booting.

(92) A Power-On Self-Test (POST) code is generated during a boot process. Server BIOS post code service **624** may be stored in private cloud server **360** and configured to communicate with RACs **314** to verify steps in a boot process.

(93) Server BIOS enumeration service **628** may be stored in private cloud server **360** and configured to communicate with RACs **314** to execute an enumeration process once a DPU OS is booted.

(94) Policy manager service **626** may store a set of policies corresponding to a boot process and a synchronization process. For a boot process, in some embodiments, policy manager service **626** may store a policy indicating a NIC OS should power on before a server OS. In some embodiments, policy manager service **626** may store a policy indicating an auxiliary-powered NIC OS and a non-auxiliary-powered NIC OS should power on before a server OS. In some embodiments, policy manager service **626** may store a policy indicating an auxiliary-powered NIC OS should power on first, then a non-auxiliary-powered NIC OS should power on, then a server OS should power on. For a shutdown process, in some embodiments, policy manager service **626** may store a policy indicating a DPU OS should power off before a server OS.

(95) Boot order/timeout configuration service **630** may be stored in private cloud server **360**. For a boot process, boot order/timeout configuration service **630** may be configured to communicate with boot synchronization service **620** and server OS halt service **636** to ensure a DPU OS is able to power up and components in server **312** receive power in an order as determined by policy manager service **626**.

(96) For a shutdown process, shutdown synchronization service **632** may be stored in private cloud server **360**. Shutdown synchronization service **632** may be configured to communicate with one or more of DPU OS halt service **634**, server OS halt service **636** and CPLD monitoring service **640** to maintain power and prevent shutdown of server OS until a DPU OS is able to shutdown safely.

(97) CPLD monitoring service **640** may manage power to components in server **312**. In some embodiments, when a graceful shutdown process is initiated, CPLD monitoring service **640** may

continue to allow components in server **312** to receive power until a DPU OS is able to safely power down.

(98) In some servers **312**, a DPU (Data processing Unit) operating system (OS) or a Network Interface Card (NIC) OS (not shown) is expected to boot first, followed by a Server OS (not shown). For example, booting a DPU OS first may ensure any DPU emulated Peripheral Component Interconnect Express (PCIe) functions are seamlessly enumerated. In some embodiments, a SmartNIC is a type of NIC that is capable of offloading some information processing from a CPU, wherein booting a SmartNIC type NIC may enable server **312** to leverage the capabilities of the SmartNIC to process information faster and/or more efficiently.

(99) When server **312** is powered off, a server OS is expected to shut down first followed by the DPU OS so that there will not be any crash. A graceful shutdown may refer to when a software function is used (e.g., a user selects a shutdown option from a menu) to turn off (power down) server **12** such that the server OS is allowed to shut down processes and close connections. In contrast, a hard shutdown may refer to when power is removed or interrupted (e.g., a user activating a power button). In some cases, if a DPU is shutdown first then the server OS will crash.

(100) Redundant Remote Access Controller Firmware

(101) Referring to FIGS. **6** and **7**, some servers **312** may contain two RACs **314** with firmware **316** in each RAC **314**. RACs **314** may communicate with each other. Referring to FIG. **7**, each RAC **314** may communicate with a BMC **702** containing a complex programming logic device (CPLD) **704**. BMCs **704** may communicate with each other over Shifty bus **706**. BMCs **704** may further communicate with co-processors **708**, wherein a secure one-to-many (e.g., $1 \times$ many) binding of the RAC firmwares **316** over I2C bus **710** to private cloud server **360** may be performed and server **312** may be marked with being in a redundancy state. In the redundancy state, embodiments may perform load balancing activities between RACs **314**. For example, if a user initiates a boot process, boot synchronization service **620** may divide the firmware update and send one set of firmware updates to one BMC **702-1** and another set of firmware updates to the other BMC **702-2**. Embodiments may maintain the consistency of a physical topology configuration among RAC firmwares **316**.

(102) However, redundant RAC firmwares **316-1** and **316-2** may complicate boot and shutdown processes. In some cases, server **312** may be configured with RACs **314-1** and **314-2** running in Active/Active mode. In such configurations where more than one RAC firmware **316** is associated with the same server **312** and the same set of server components, there may be conflicts. Furthermore, in some Computer-Mediate Communication (CMC) platforms, there may be redundant embedded controller (EC) firmware running in Active/Passive mode wherein redundant EC firmware may be activated as a primary EC firmware.

(103) In some embodiments, server **312** may have a dedicated witness node **610**. Witness node **610** may be configured to detect failover events and is supposed to be powered on before other components in server **312**.

(104) Boot synchronization and shutdown synchronization may be based on the system configuration (e.g., with or without DPU, an Auxiliary-Powered DPU or a Non-Auxiliary-Powered DPU) and a topology (RACs **314** configured in an Active/Active mode or an Active/Passive mode).

(105) Referring to FIG. **8**, embodiments of a method for synchronizing a boot process may ensure a component OS (e.g., a DPU OS or SmartNIC OS) boots first before server OS. Flow diagram **800** may begin at step **802** when server **312** determines a user is attempting to power on server **312**. Embodiments may power on all the components together but ensure that Server BIOS is waiting until any component operating systems are booted and then the server OS is booted.

(106) At step **804**, boot synchronization service **620** may communicate a signal to an auxiliary-powered NIC indicating server **312** is receiving power. In some embodiments, auxiliary-powered NICs may remain always on, wherein boot synchronization service **620** may communicate a signal to auxiliary-powered NIC OS indicating alternating current (AC) power is on.

(107) At step **806**, auxiliary-powered NIC OS may communicate a signal to non-auxiliary-powered NIC OS indicating server **312** is receiving power. In some embodiments, non-auxiliary powered NICs may be powered off when server **312** is powered off, wherein an auxiliary-powered NIC OS may communicate a signal to a non-auxiliary-powered NIC OS indicating AC power is on.

(108) At step **808**, non-auxiliary-powered NIC OS may communicate a signal to server BIOS/OS/witness node **610** indicating server **312** is receiving power.

(109) At step **810**, boot synchronization service **620** may communicate a signal to server BIOS/OS/witness node **610** to enable server BIOS waiting, wherein server BIOS/OS/witness node **610** waits (e.g., does not continue with a boot process).

(110) At step **812**, boot synchronization service **620** may wait for components to finish booting. For example, boot synchronization service **620** may wait for a non-auxiliary-powered NIC OS and an auxiliary-powered NIC OS to finish booting.

(111) At step **814**, a component may communicate a signal to boot synchronization service **620** indicating the component has finished a boot process. For example, a non-auxiliary-powered NIC may communicate a signal to boot synchronization service **620** indicating the non-auxiliary-powered NIC OS has finished booting.

(112) At step **816**, a component may communicate a signal a signal to boot synchronization service **620** indicating the component has finished a boot process. For example, an auxiliary-powered NIC may communicate a signal to boot synchronization service **620** indicating the auxiliary-powered NIC OS has finished booting.

(113) At step **818**, boot synchronization service **620** may communicate a signal to server BIOS/OS/witness node **610** to disable BIOS waiting, wherein server BIOS/OS/witness node **610** may finish powering up.

(114) Referring to FIG. **9**, embodiments of a method for synchronizing a shutdown process may ensure the server OS shuts down first followed by the DPU OS so that there will not be any crash. Flow diagram **900** depicts steps to shut down all component operating systems together but ensure that a server OS is shutdown only after all the component operating systems are shutdown.

(115) Flow diagram **900** may begin at step **902** when a user activates a graceful shutdown process on server **312**.

(116) At step **904**, shutdown synchronization service **632** may communicate a signal to server BIOS/OS node **610** indicating server BIOS/OS node **610** should begin a shutdown process.

(117) At step **906**, shutdown synchronization service **632** may communicate a signal to CPLD monitoring service **640** indicating CPLD monitoring service **640** should hold or maintain power to server **312**.

(118) At step **908**, server BIOS/OS node **610** initiates a shutdown process.

(119) At step **910**, shutdown synchronization service **632** may communicate a signal to DPU BIOS/OS node **612** indicating DPU BIOS/OS node **612** should begin a shutdown process.

(120) At step **912**, DPU BIOS/OS node **612** initiates the shutdown process.

(121) At step **914**, shutdown synchronization service **632** may communicate a request for a DPU OS halt status to DPU BIOS/OS node **612**.

(122) At step **916**, DPU BIOS/OS node **612** may communicate a response to shutdown synchronization service **632**.

(123) At step **918**, shutdown synchronization service **632** may wait until DPU BIOS/OS node **612** communicates an OS halt status indicating the DPU OS is halted (e.g., not processing information).

(124) At step **920**, shutdown synchronization service may communicate a signal to CPLD monitoring service **640** indicating it is safe to power off server **312**.

(125) At step **922**, CPLD monitoring service **640** may remove the main power from server **312**.

(126) Thus, embodiments may comprise server **312** communicatively coupled to private cloud server **360** storing a set of services for communicating with one or more RACs **314**, a server processor executing instructions in a BIOS/OS node **610** and a component node (e.g., DPU

BIOS/OS node **612**) to synchronize a boot process, which include booting a server OS and a component OS based on a boot order specified in a policy manager service **626**.

(127) Furthermore, embodiments may comprise server **312** communicatively coupled to private cloud server **360** storing a set of services for communicating with one or more RACs **314**, a server processor executing instructions in OS node **610** and a component node (e.g., DPU BIOS/OS node **612**) to synchronize a shutdown process, which include powering down a Server OS and a component OS based on a shutdown order specified in a policy manager service **626**.

(128) Embodiments allow for easy customization and configuration to accommodate new components (e.g., DPUs, NICs) in servers **312** and accommodate topology-based device synchronization.

(129) The above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments which fall within the true spirit and scope of the present disclosure. Thus, to the maximum extent allowed by law, the scope of the present disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

Claims

1. A system comprising: a server comprising: a plurality of components, each component storing a component operating system (OS); a server memory storing: a server OS; a component node for each component OS; and a remote access controller (RAC); and a private cloud server (PCS) communicatively coupled to the server, the private cloud server comprising: a PCS processor; and a PCS memory storing: a boot synchronization service executable by the PCS processor to: communicate with the RAC to identify the plurality of components; determine a boot order for the plurality of components; and communicate with the RAC to complete, based on the boot order, a boot process for the plurality of components before completing a boot process for the server OS; a shutdown synchronization service configured to: determine a shutdown order for the plurality of components; and communicate with the RAC to complete, based on the shutdown order, a shutdown process for the plurality of components before completing a shutdown process for the server OS.
2. The system of claim 1, wherein the PCS memory stores: a policy manager service storing a set of policies; and a boot order/timeout configuration service configured to determine the boot order based on a policy stored in the policy manager service, wherein the boot synchronization service is configured to communicate with the boot order/timeout configuration service to determine the boot order.
3. The system of claim 2, wherein the plurality of components comprises a data processing unit (DPU); and the PCS memory stores a DPU OS halt service executable by the PCS processor to communicate with a DPU Basic Input/Output System (BIOS)/OS node to determine when the boot process for the DPU OS is completed.
4. The system of claim 2, wherein the PCS memory stores a DPU monitoring service executable by the PCS processor to halt the shutdown process for the server OS until the shutdown process for the plurality of components is completed.
5. The system of claim 1, wherein the PCS memory stores a server OS halt service executable by the PCS processor to communicate with a server Basic Input/Output System (BIOS) node to halt the boot process for the server OS until the boot process for the plurality of components is completed.
6. The system of claim 5, wherein the PCS memory stores a complex programmable logic device (CPLD) monitoring service executable by the PCS processor to maintain a power supply to the server until the boot process for the plurality of components is completed.

7. A data center comprising: a plurality of servers, each server comprising: a plurality of components, each component storing a component operating system (OS); a server memory storing: a server OS; a component node for each component OS; and a remote access controller (RAC); and a private cloud server (PCS) communicatively coupled to the plurality of servers, the private cloud server comprising: a PCS processor; and a PCS memory storing: a boot synchronization service executable by the PCS processor to: communicate with the RAC in a server of the plurality of servers to identify the plurality of components; determine a boot order for the plurality of components; and communicate with the RAC to complete, based on the boot order, a boot process for the plurality of components before completing a boot process for the server OS; a shutdown synchronization service configured to: determine a shutdown order for the plurality of components; and communicate with the RAC to complete, based on the shutdown order, a shutdown process for the plurality of components before completing a shutdown process for the server OS.
8. The data center of claim 7, wherein the PCS memory stores: a policy manager service storing a set of policies; and a boot order/timeout configuration service configured to determine the boot order based on a policy stored in the policy manager service, wherein the boot synchronization service is configured to communicate with the boot order/timeout configuration service to determine the boot order.
9. The data center of claim 8, wherein the plurality of components comprises a data processing unit (DPU); and the PCS memory stores a DPU OS halt service executable by the PCS processor to communicate with a DPU Basic Input/Output System (BIOS)/OS node to determine when the boot process for the DPU OS is completed.
10. The data center of claim 8, wherein the PCS memory stores a DPU monitoring service executable by the PCS processor to halt the shutdown process for the server OS until the shutdown process for the plurality of components is completed.
11. The data center of claim 7, wherein the PCS memory stores a server OS halt service executable by the PCS processor to communicate with a server Basic Input/Output System (BIOS) node to halt the boot process for the server OS until the boot process for the plurality of components is completed.
12. The data center of claim 11, wherein the PCS memory stores a complex programmable logic device (CPLD) monitoring service executable by the PCS processor to maintain a power supply to the server until the boot process for the plurality of components is completed.
13. A method comprising: communicating, by a private cloud server (PCS) processor with a remote access controller (RAC) of a server, to identify a plurality of components of the server, each component storing a component operating system (OS); detecting, by the private cloud server (PCS) processor, a boot order for the plurality of components of the server; communicating, by the PCS processor with the RAC of the server, to complete, based on the boot order, a boot process for the plurality of components before completing a boot process for a server operating system (OS) of the server; determining a shutdown order for the plurality of components; and communicating, by the PCS processor with the RAC of the server, a signal to complete, based on the shutdown order, a shutdown process for the plurality of components before completing a shutdown process for the server OS.
14. The method of claim 13, further comprising: storing a set of policies in a policy manager service; and determining, by a boot order/timeout configuration service, the boot order based on a policy stored in the policy manager service, wherein the boot synchronization service is configured to communicate with the boot order/timeout configuration service to determine the boot order.
15. The method of claim 14, wherein the plurality of components comprises a data processing unit (DPU); and the method further comprises communicating, by a DPU OS halt service stored in the PCS memory, with a DPU Basic Input/Output System (BIOS)/OS node to determine when the boot process for the DPU OS is completed.

16. The method of claim 14, further comprising a DPU monitoring service halting the shutdown process for the server OS until the shutdown process for the plurality of components is completed.
 17. The method of claim 13, further comprising a server OS halt service halting the boot process for the server OS until the boot process for the plurality of components is completed.
 18. The method of claim 17, further comprising a complex programmable logic device (CPLD) monitoring service maintaining a power supply to the server until the boot process for the plurality of components is completed.
-